Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI engineer and production systems builder with 8+ years shipping regulated-data platforms at enterprise scale (100K+ MAU, 1.6M regulated records, 99.5% uptime). Built **Leasel** - an 8-agent LangGraph platform for regulated compliance operations with RAG retrieval, confidence scoring, and reproducible eval harness (93.6% precision across 1,240 enterprise documents). Research Associate at George Mason University building **predictive models on WHO health datasets across 190+ countries** (R^2=0.977) and Tableau dashboards supporting clinical research leadership. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI & LangGraph | Multi-Agent Orchestration | RAG Pipelines | Eval Harnesses & LLM Evaluation | NLP & Document Intelligence | Clinical & Health Data Analytics | Predictive Analytics | PHI & Compliance Engineering | Production AI Systems | Regulated Data Pipelines

WORK EXPERIENCE

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant

- Delivered technical workshops to 600+ students - Python programming, AI fundamentals, database design; one-on-one technical advisory and debugging support across two subject areas.

George Mason University - Research

Aug 2024 - May 2025

Research Associate - AI & Predictive Analytics

- Built **multi-source predictive models on WHO health datasets** (Python, R) across 190+ countries - R^2=0.977; applied Isolation Forest anomaly detection surfacing data quality issues manual review missed across 3 research pipelines; quantitative analysis driving faculty research strategy.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - 40% research comprehension improvement; translated complex multi-dimensional health data into stakeholder-ready visualizations.

Magizhs Tech Solution

May 2025 - Aug 2025

AI Engineer (Intern)

- Architected and deployed a LangGraph agentic system with RAG retrieval, confidence-scored outputs, and validation guardrails - self-directed from requirements through production across three commercial verticals.

Entrata - PropTech / Real Estate SaaS (100K+ MAU)

Sep 2018 - Aug 2024

Senior Software Engineer

- **Shipped 20+ production modules** (PHP, Python, React, PostgreSQL, RabbitMQ, AWS) across 120+ Agile sprints for 100K+ monthly active users - 99.5% uptime sustained across 6+ consecutive months.
- **Built regulated data pipelines** on 1.6M financial records - HUD, LIHTC, HCV compliance - with output validation gates, data quality controls, and audit trails; zero compliance violations across 7 years.
- **100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines; optimized 200+ SQL queries (30% performance gain); eliminated 80% of manual workload via Python/Bash automation.
- Led microservices redesign (Docker) - 50% scalability improvement; mentored 5+ engineers across 300+ peer code reviews.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines on 1.6M regulated records; AWS S3 cloud migration.

PROJECTS

Leasel: 8-Agent AI Platform for Regulated Compliance Operations 2026

Production agentic system: 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Escalation) with RAG retrieval (91% Precision@1, 100% citation correctness) and confidence scoring. Reproducible eval harness with LLM-as-judge scoring, pass@k multi-trial sampling, and regression gating - validates agents in simulation before every release. Production precision: 93.6% across 1,240 enterprise documents.

Python · LangGraph · FastAPI · RAG · Confidence Scoring · LLM-as-Judge · Eval Harness · Regulated Data

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University 2017

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, PHP, R, Bash

AI: LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, NLP, LLMs, BERT, spaCy, Embeddings, Document Intelligence, Confidence Scoring, LLM-as-Judge, Eval Harnesses, Agentic AI, FastAPI, ChromaDB, OpenAI API, Anthropic API, Predictive Analytics, Health Data Modeling, Statistical Modeling

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, REST APIs, Git, Linux, Agile/Scrum

Data & ML: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Pandas, NumPy, Tableau, Power BI, ggplot2, Quarto